

COMPUTER PROGRAMMING CONCEPTS (390)

—OPEN EVENT—

NATIONAL – 2019

DO NOT WRITE ON TEST BOOKLET

TOTAL POINTS

_____ (100 points)

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than sixty (60) minutes testing time

Property of Business Professionals of America.
May be reproduced only for use in the Business Professionals of America
Workplace Skills Assessment Program competition.

Identify the choice that best completes the statement or answers the question. Mark “A” if the statement is true. Mark “B” if the statement is false.

1. Which of the following functions substitutes a sequence of characters in a string variable with another sequence of characters?
 - a. change
 - b. replace
 - c. modify
 - d. no such function
2. Which of the following arguments is *optional* for the creation of input and output objects?
 - a. open
 - b. close
 - c. eof
 - d. mode
3. In OOP, making data private is providing _____.
 - a. redundancy
 - b. inheritance
 - c. encapsulation
 - d. polymorphism
4. Stream manipulators that do *not* have arguments, such as *fixed* and *scientific*, are defined in which file?
 - a. namespace
 - b. iostream
 - c. iomanip
 - d. none of the above
5. What type of functions or methods do *not* return a value after completing their task?
 - a. value-defined
 - b. loop
 - c. all
 - d. void
6. How do you pass a variable when you want the receiving function to change the contents of the original variable?
 - a. by value
 - b. by reference
 - c. by address
 - d. by email

7. In some languages, what must a program contain to use the *string* class?
 - a. `#include <string.h>`
 - b. `import java.lang`
 - c. Either a or b
 - d. None of the above
8. The function or method header for a value-returning function or method begins with which of the following?
 - a. opening and closing parenthesis
 - b. opening braces
 - c. the keyword function or method
 - d. the *returnDataType*
9. What is the requirement for *not* repeating the instructions in a repetition structure called?
 - a. nesting
 - b. loop exit condition
 - c. sentinel value
 - d. looping condition
10. What is a factor that remains the *same* throughout a program?
 - a. constant
 - b. variable
 - c. algorithm
 - d. object
11. Which of the following types of loops is often used in programs that allow the user to select from a menu?
 - a. pretest
 - b. posttest
 - c. *for*
 - d. *while*
12. When declaring and initializing an array, which of the following *best* describes the status of the *initialValues* section?
 - a. contained within quotes
 - b. contained within brackets
 - c. optional
 - d. required
13. What type of decision is always made by an outer selection structure?
 - a. alternative
 - b. outer
 - c. secondary
 - d. primary
14. A base class is created from a derived class.
 - a. true
 - b. false

15. What is the expected output of the following switch statement?

```
switchval = 81-21
switch (switchval)

    case 30:    print "switchval is 30"
                break
    case 40:    print "switchval is 40"
                break
    case 50:    print "switchval is 50"
                break
    default:    print "switchval is more than 50"
```

- a. switchval is 30
 - b. switchval is 40
 - c. switchval is 50
 - d. switchval is more than 50
16. In some languages, what must a program contain to use the *random* class?
- a. #include <cstdlib>
 - b. import java.util.lang
 - c. Either a or b
 - d. None of the above
17. What is the human-readable version of a program created in a high-level language by a programmer called?
- a. object code
 - b. op code
 - c. structured English
 - d. source code
18. Which programming language is *not* used for artificial intelligence applications and expert systems?
- a. COBOL
 - b. Prolog
 - c. C++
 - d. Objective C
19. A programmer determines whether a problem's solution requires a nested selection structure by studying which of the following?
- a. flow model
 - b. data validation tables
 - c. problem specification
 - d. none of the above
20. With parallel arrays, both arrays must contain elements of the same data type.
- a. true
 - b. false

21. What is the end result of the following code?

```
int xNum = 0
int yNum = 0
do
    xNum = xNum + yNum
    yNum += 1
    print "xNum: " + xNum
while (xNum > 2)
```

- a. xNum: 2 is printed
- b. xNum: 0 is printed
- c. endless loop
- d. no output, the loop is never executed

22. What is the result of the following arithmetic expression in most languages?

$2 + 4 * 2$

- a. 8
- b. 12
- c. 10
- d. 16

23. In the following code, what is the scope and lifetime of the *num1* variable?

```
int num2 = 100

int main( )
    for (int num1 = 1; num1 < num2; num1 += 1)
        .....more code here.....
```

- a. global, until the for loop ends
- b. local, until the for loop ends
- c. global, until the program ends
- d. local, until the program ends

24. Given the following array declaration, what is the value stored in the scores[1][1] element?

```
int scores[3][3] = {{1,2,3}}
```

- a. 0
- b. 1
- c. 2
- d. 3

25. Which *string* class function or method would be used to remove one character located anywhere in a *string* variable?

- a. remove
- b. erase
- c. a or b
- d. None of the above

26. What is the output of the code below?

```
int x = 25
int y = 8
print (x + y) / y)
```

- a. 0
- b. 3
- c. 4.125
- d. 4

27. What is the output of the code below?

```
int huey = -10
int dewey = huey * 3
int louie = dewey - 3
print (louie %= huey)
```

- a. -3.0
- b. -3
- c. 3
- d. 3.3

28. What diagram would be used to represent class?

- a. flowchart
- b. visualization document
- c. UML
- d. none of the above

29. What is the standard computer language for relational database management and data manipulation?

- a. UML
- b. DBQ
- c. PHP
- d. SQL

30. Which of the following are referred to as the arguments that are passed to a function or method?

- a. corresponding arguments
- b. required arguments
- c. formal arguments
- d. actual arguments

31. What is assigned to *newStr* after the following code is executed?

```
string orgStr = "Hello there!"  
string newStr = orgStr.substr ( 1, orgStr.length( ) - 1)
```

- a. Hello there
- b. ello there!
- c. ello there
- d. Hello there!

32. What type of array resembles a table with rows and columns?

- a. one-dimensional
- b. scalar
- c. integer
- d. two-dimensional

33. What is (are) the argument(s) passed to the *pow* function or method?

- a. the dividend and divisor
- b. base number and exponent
- c. the number to be squared
- d. a long value

34. Which of the following SQL statements selects all values from the "Country" column in the "Customers" data table?

- a. SELECT Customers FROM Country
- b. FROM Customers SELECT Country
- c. SELECT Country FROM Customers
- d. FROM Country SELECT Customers

35. In OOP terminology, when a class is created from another class, what is the original class called?

- a. original
- b. derived
- c. child
- d. base

36. Which of the following terms describes the process of connecting or linking strings together?

- a. string concatenation
- b. string constructing
- c. string linkage
- d. string meshing

37. What does the term UML stand for?
- a. United Modeling Language
 - b. Unified Mobility Language
 - c. Unified Modeling Language
 - d. United Mobility Language
38. In OOP terminology, when a class is created from another class, what is the newly created class called?
- a. original
 - b. derived
 - c. parent
 - d. base
39. What is SQL used for besides to query data?
- a. insert data
 - b. modify data
 - c. update data
 - d. All of the above
40. Which of the following SQL statements selects only the distinct values from the “Country” column in the “Customers” data table?
- a. SELECT Customers FROM DISTINCT Country
 - b. SELECT DISTINCT Country FROM Customers
 - c. FROM Customers SELECT DISTINCT Country
 - d. FROM DISTINCT Country SELECT Customers
41. Which of the following SQL statements lists the number of different (distinct) customer “Countries” from the “Customers” data table?
- a. SELECT Customers FROM COUNT (DISTINCT Country)
 - b. FROM Customers SELECT COUNT (DISTINCT Country)
 - c. FROM COUNT (DISTINCT Country) SELECT Customers
 - d. SELECT COUNT (DISTINCT Country) FROM Customers
42. Which of the following SQL statements selects all the customers from the country “Mexico”, in the “Customers” data table?
- a. SELECT * FROM Customers WHERE Country = ‘Mexico’
 - b. SELECT Customers FROM * WHERE Country = ‘Mexico’
 - c. FROM customers SELECT * WHERE Country = ‘Mexico’
 - d. FROM * SELECT Customers WHERE Country = ‘Mexico’
43. Which of the following SQL statements selects all fields from “Customers” where country is “Germany” AND city is “Berlin”, in the “Customers” data table?
- a. SELECT Customers FROM * WHERE Country = ‘Germany’ AND City = ‘Berlin’
 - b. FROM customers SELECT * WHERE Country = ‘Germany’ AND City = ‘Berlin’
 - c. SELECT * FROM Customers WHERE Country = ‘Germany’ AND City = ‘Berlin’
 - d. FROM * SELECT Customers WHERE Country = ‘Germany’ AND City = ‘Berlin’

44. Which of the following SQL statements selects all fields from “Customers” where country is NOT “Germany”, in the “Customers” data table?
- a. SELECT Customers FROM * WHERE NOT Country = ‘Germany’
 - b. FROM customers SELECT * WHERE NOT Country = ‘Germany’
 - c. FROM * SELECT Customers WHERE NOT Country = ‘Germany’
 - d. SELECT * FROM Customers WHERE NOT Country = ‘Germany’
45. What subscript should be used to access the fourth element in a one-dimensional array?
- a. 2
 - b. 3
 - c. 4
 - d. 5
46. Which of the following shows the subscript of where the elements located in the first row in a two-dimensional array are assigned?
- a. 0
 - b. 1
 - c. 2
 - d. None of the above
47. What does a programmer use to tell the compiler to add the contents of a header file in the program?
- a. a public declaration
 - b. a private declaration
 - c. an include directive
 - d. a using statement
48. Which of the following would be used when specifying an array element in a two-dimensional array?
- a. [column, row]
 - b. [row, column]
 - c. a or b
 - d. None of the above
49. What is the *first* step in the problem-solving process?
- a. plan the algorithm
 - b. write the documentation
 - c. code the algorithm into a program
 - d. analyze the problem
50. What is the *second* step in the problem-solving process?
- a. plan the algorithm
 - b. write the documentation
 - c. code the algorithm into a program
 - d. analyze the problem